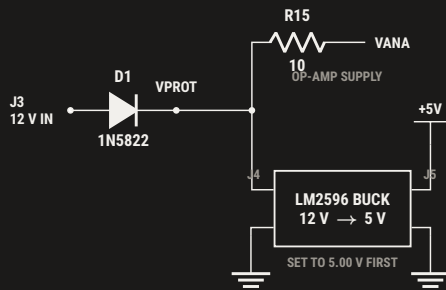


2.4 GHZ FMCW RADAR | BREADBOARD AND HARNESS – SCHEMATIC

STAGE 1 · ONE TX HORN, ONE RX HORN · RADAR-DEV

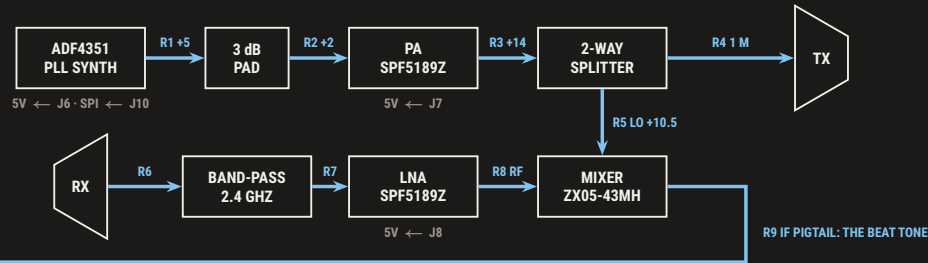
POWER

12 V IN, PROTECTED, SPLIT INTO VANA AND 5 V



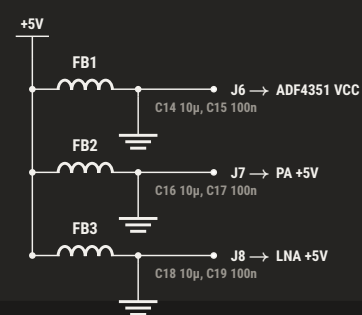
RF CHAIN

BOUGHT MODULES ON THE PLYWOOD, JOINED BY SMA COAX (BLUE). NUMBERS ARE SIGNAL LEVELS IN dBm.



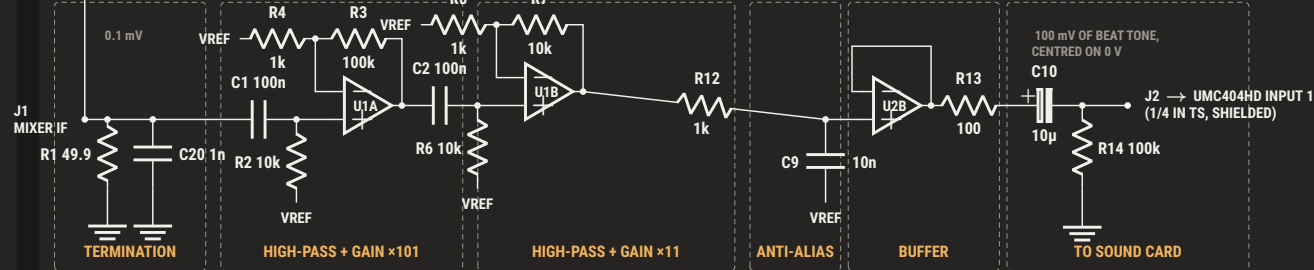
5 V TO THE RF MODULES

ONE FERRITE BEAD + 10μ + 100n EACH



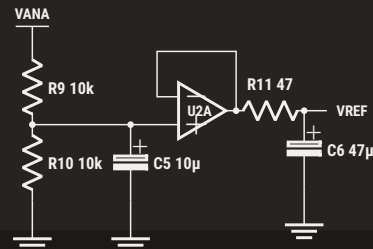
BEAT-TONE AMPLIFIER

TOTAL GAIN 1111 (61 dB), PASSES 160 HZ – 15.9 KHZ. C1, C2 MUST BE FILM.

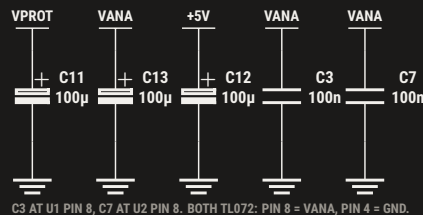


HALF-RAIL REFERENCE

VREF = VANA ÷ 2 ≈ 5.8 V: THE AMPLIFIER'S "ZERO"

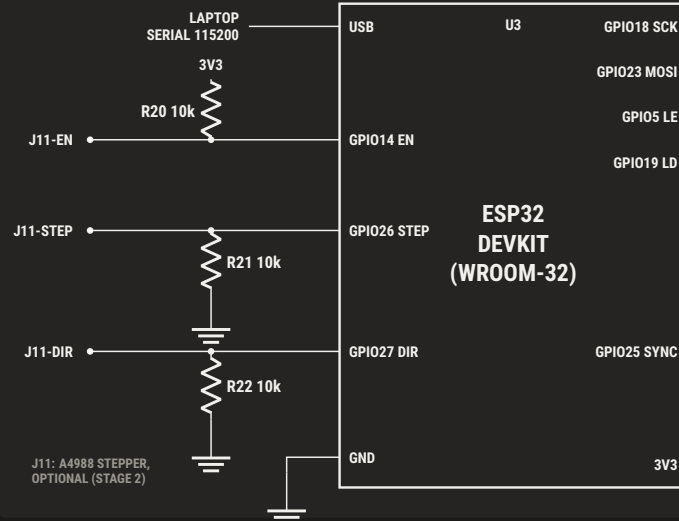


DECOUPLING



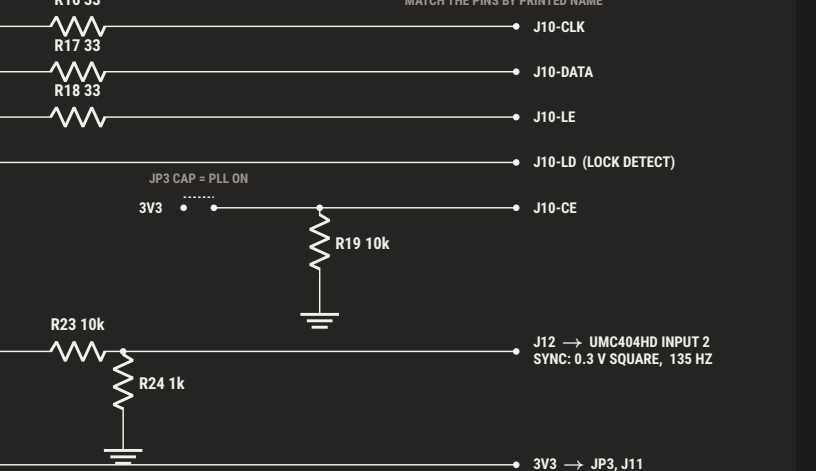
CONTROL: ESP32 RUNS RADAR_CTL

STEPS THE PLL THROUGH THE SWEEP AND MARKS EACH SWEEP ON SYNC. R16–R18 SOFTEN THE CLOCK EDGES.



ADF4351 LOGIC (J10)

MATCH THE PINS BY PRINTED NAME



J = HEADER ON THE BREADBOARD (OPEN CIRCLE = WIRE LEAVES THE BOARD) · BLUE = SMA COAX · GROUNDS ALL MEET AT THE STAR TERMINAL